



P.E.S. College of Engineering, Mandya-571401
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Department of Information Science and Engineering

Project presentation on :

**“SignAI: Indian Sign Language Recognition Using VideoMAE-
Based Vision Transformers”**

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Introduction

- Sign language plays an important role in communication for people with hearing and speech disabilities.
- Existing gesture recognition systems face difficulties in understanding continuous and dynamic sentence-level gestures.
- This project proposes **SignAI**, a transformer-based Indian Sign Language recognition system using the **VideoMAE** model.
- The system converts gesture videos into meaningful text and speech output through sequence-based deep learning and an integrated frontend-backend pipeline.

Contribution to SDG 9: Industry, Innovation, and Infrastructure

The project uses advanced AI technologies such as computer vision and VideoMAE to develop an innovative communication system.

SDG 10 – Reduced Inequalities

The system helps reduce communication barriers between hearing-impaired individuals and others, promoting equal opportunities and inclusive communication in society.

Research Gap

- Existing systems mainly focus on static or isolated gesture recognition.
- Traditional CNN and CNN-LSTM models struggle to capture long-range temporal dependencies.
- YOLO-based approaches focus more on hand detection than complete gesture understanding.
- Many existing methods lack an end-to-end system with frontend, backend, and real-time text/speech output integration.

Problem Statement



Communication between hearing-impaired individuals and normal speakers is difficult because most people do not understand Indian Sign Language (ISL). Existing systems mainly recognize isolated gestures and fail to understand continuous sentence-level gestures effectively. Therefore, an efficient sequence-based sign language recognition system is needed to convert gesture videos into meaningful text and speech output.

Project Objective

1. **To preprocess** ISL video data into structured frame sequences for effective learning
2. **To develop** a VideoMAE-based model for recognizing sentence-level sign language gestures
3. **To build** an integrated system combining frontend interface and backend model inference

Proposed Methodology

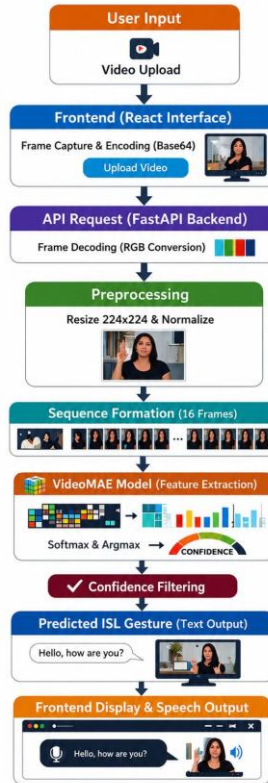


Figure 1 : Architecture of SIGNAL

Objective 1

Goal:

- Convert raw ISL videos into structured sequences for gesture understanding

Methodology:

- Extract frames from video input
- Resize to 224×224 and normalize
- Arrange frames in temporal order
- Form fixed-length sequences (16 frames)

Implementation:

- Processed ISL-CSLTR dataset (~7,671 videos, 101 classes)
- Generated sequence-based dataset for model training
- Ensured consistency across all gesture samples

Objective 2

Goal:

- Develop a robust model to recognize sentence-level ISL gestures

Methodology:

- Use pretrained VideoMAE (Transformer-based model)
- Fine-tune on ISL gesture dataset
- Learn spatial + temporal features using self-attention
- Perform multi-class classification (Softmax)

Implementation:

- Trained on sequence data (16-frame input)
- Used GPU for training (~3 days)
- Generated gesture predictions with confidence score

Objective 3

Goal:

- Develop a complete system to convert gesture videos into text output

Methodology:

- Capture input through video upload
- Process frames and send to backend
- Perform model inference using VideoMAE
- Display predicted gesture as text output

Implementation:

- Frontend: React-based user interface
- Backend: FastAPI for model inference
- Integrated end-to-end pipeline (Input → Prediction → Output)

Output

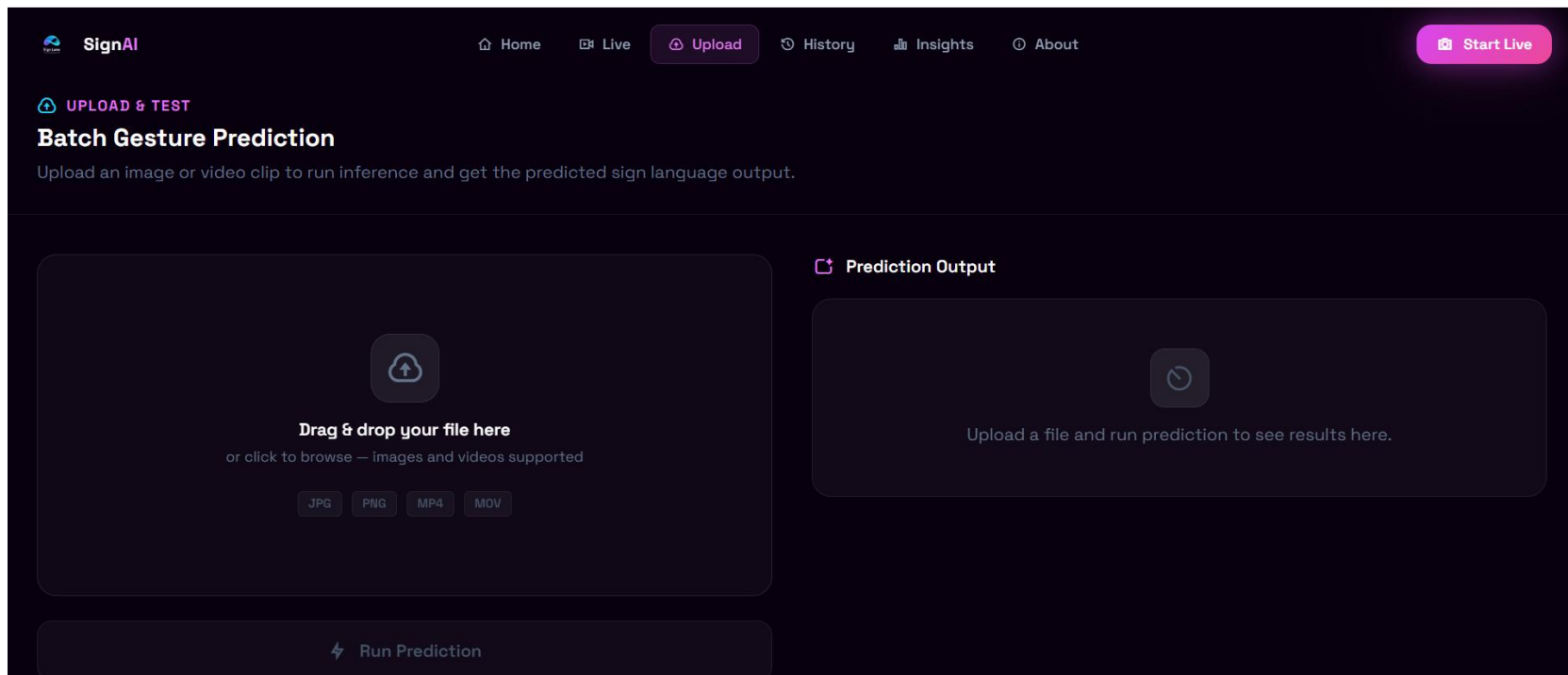


Fig 1: Video Upload Interface

Output

The screenshot displays the SignAI web application interface. The top navigation bar includes links for Home, Live, Upload, History, Insights, and About, along with a Start Live button. The main content area is divided into two panels. The left panel, titled 'Uploaded File', shows a video player with a woman in a patterned sari. Below the video, the file is identified as 'food (4).MP4' with a size of 13.80 MB. The right panel, titled 'Prediction Output', shows a loading spinner and the text 'Analyzing gesture...' followed by 'Running VideoMAE Transformer Inference'. A progress list indicates the steps: Preprocessing frame, Extracting spatial features, Sequence inference, and Post-processing. At the bottom, a status bar shows 'Running Inference...'.

SignAI

Home Live Upload History Insights About Start Live

Uploaded File Remove

0:00 / 0:04

food (4).MP4 13.80 MB

Prediction Output

Analyzing gesture...

Running VideoMAE Transformer Inference

- Preprocessing frame
- Extracting spatial features
- Sequence inference
- Post-processing

Running Inference...

Fig 2: Model Inference Process

Output

The screenshot displays the SignAI web application interface. At the top, the navigation bar includes links for Home, Live, Upload, History, Insights, and About, along with a Start Live button. The main content area is divided into two sections. On the left, the 'Uploaded File' section shows a video player with a woman in a patterned sari, a progress bar at 0:00 / 0:04, and file details: 'food (4).MP4' and '13.80 MB'. A 'Remove' button is also present. On the right, the 'Prediction Output' section displays the detected gesture as 'had your food' in large, stylized text. Below this, a 'Confidence Score' of 99.9% is shown with a corresponding progress bar. A 'Text Output' box also displays 'had your food'. At the bottom of the prediction section is a 'Play Voice Output' button. A large blue 'Run Prediction' button is located at the bottom center of the interface.

SignAI

Home Live Upload History Insights About Start Live

Uploaded File Remove

0:00 / 0:04

food (4).MP4 13.80 MB

Run Prediction

Prediction Output

DETECTED GESTURE

had your food

Confidence Score 99.9%

Text Output

had your food

Play Voice Output

Fig 3 : Gesture Prediction Output

Performance Evaluation

Evaluation Process

- Compared CNN, CNN+LSTM, YOLO, and VideoMAE models
- Analyzed training and validation performance
- Studied confusion matrix and prediction behaviour
- Evaluated model robustness using graphs and metrics

Results

- VideoMAE achieved 74% validation accuracy
- YOLO achieved higher detection accuracy (85%)
- VideoMAE showed better temporal understanding of gestures
- Transformer-based learning improved sequence recognition performance

Performance Comparison with Existing Methods

Results

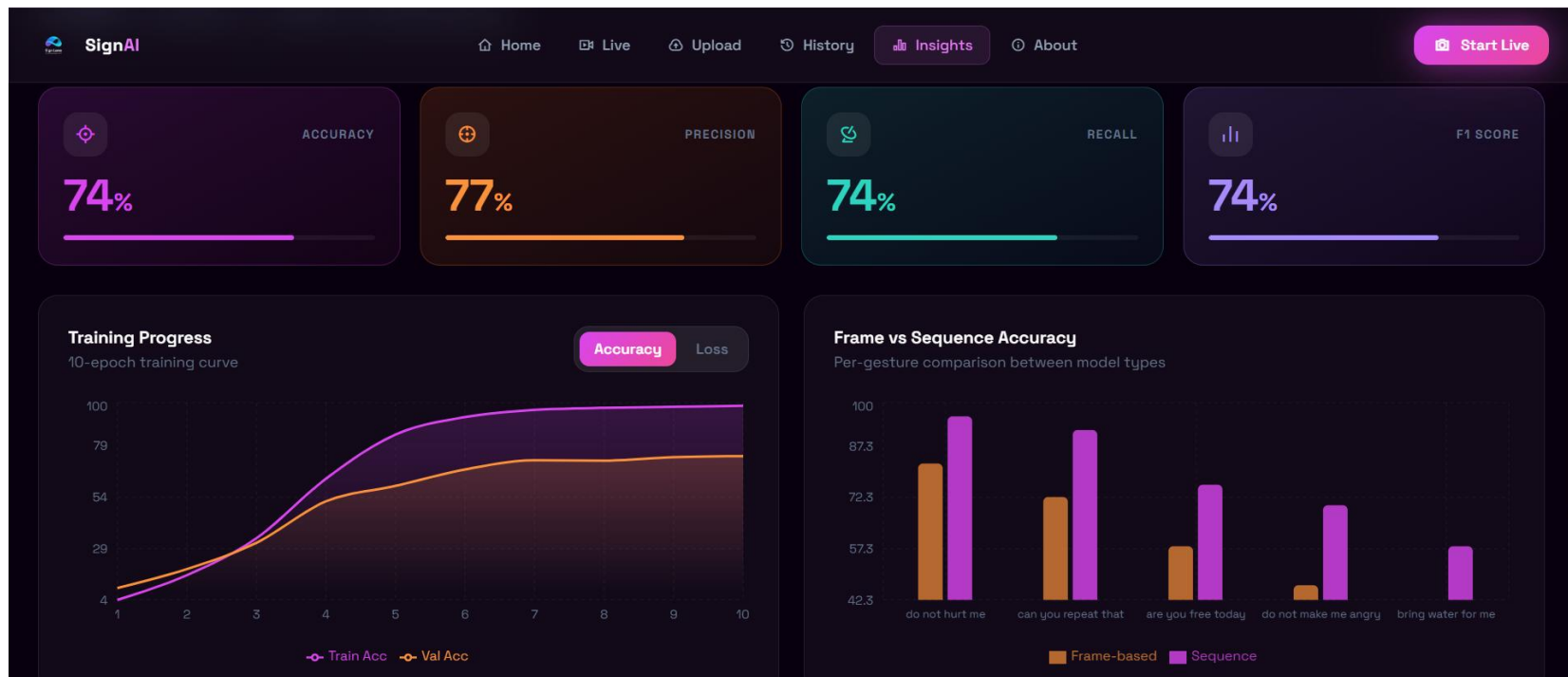


Fig 4 : Model Performance Metrics

Comparative Analysis

Model	Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC
CNN-Based	Spatial	65	67	64	65	0.85
CNN + LSTM	Hybrid	69	70	68	69	0.88
YOLO-Based	Detection	85	87.5	85	85	0.90
Proposed VideoMAE (SignAI)	Transformer	74	77	74	74	0.98

Table 1: Performance Comparison with Existing Methods

Future Scope and Applications of SignAI

Future Work

- Implement real-time webcam-based gesture recognition
- Expand the dataset with more ISL sentence-level gestures
- Improve model accuracy using advanced transformer architectures
- Develop a mobile application for portable accessibility support
- Integrate multilingual text and speech translation features

Applications of the Proposed System

- Assistive communication tool for deaf and mute individuals
- Smart classrooms and inclusive learning environments
- Public service and customer support assistance systems
- AI-based accessibility solutions in healthcare and workplaces
- Real-world human-computer interaction and communication platforms

Tools and Languages

Programming & Frameworks

- Python
- React.js
- FastAPI

Deep Learning & Libraries

- PyTorch
- Transformers (VideoMAE)
- OpenCV

Model & Techniques

- VideoMAE (Transformer-based Model)
- Sequence-based Processing (16 frames)
- Softmax Classification

Development & Execution

- GPU (Model Training – ~3 days)
- CPU (Inference & Deployment)

Dataset

- ISL-CSLTR Dataset (101 Classes)

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Thank You